

GUELPH TURFGRASS INSTITUTE, ON, Canada

WMO: 718330

Lat: 43.5472N

Lon: 80.2151W

Elev: 325

StdP: 97.48

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.5	-18.5	-24.8	0.4	-20.7	-22.0	0.5	-18.1	9.4	-6.0	8.6	-6.8	2.0	70	0.544

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	29.7	21.5	28.2	20.8	26.7	20.0	23.1	27.5	22.2	26.3	21.2	25.0	3.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	17.1	25.3	20.8	16.1	24.3	19.9	15.2	23.6	70.3	27.4	66.7	26.4	63.2	25.1	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.5	6.6	DB	-25.6	31.9	4.0	1.8	-28.5	33.1	-30.8	34.2	-33.1	35.2	-35.9	36.5	(4)
(5)			WB	-25.6	25.0	4.2	1.6	-28.6	26.2	-31.0	27.1	-33.4	28.0	-36.5	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-7.3	-6.5	-1.6	5.1	11.7	17.3	19.3	18.6	14.8	8.5	2.4	-2.9
	DBStd	10.52	6.43	5.99	6.14	4.75	4.52	3.78	3.14	2.97	4.05	4.43	4.52	5.21
	HDD10.0	2272	536	462	363	159	32	1	0	0	6	84	229	401
	HDD18.3	4398	794	695	617	397	211	64	26	34	120	305	477	659
	CDD10.0	1066	0	0	4	13	86	219	289	265	150	38	2	0
	CDD18.3	151	0	0	0	0	6	32	56	41	14	1	0	0
	CDH23.3	1503	0	0	1	4	83	344	539	368	157	7	0	0
	CDH26.7	326	0	0	0	0	11	87	126	69	32	1	0	0
Wind	WSAvg	3.0	3.6	3.6	3.6	3.7	3.1	2.5	2.3	2.1	2.2	2.8	3.2	3.5
Precipitation	PrecAvg	905	74	60	60	77	84	83	86	77	78	79	80	69
	PrecMax	1193	121	119	139	128	159	196	154	161	141	140	154	128
	PrecMin	670	44	29	21	31	32	24	27	23	25	20	30	34
	PrecStd	130	23	24	29	30	28	40	35	33	31	33	30	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	10.7	20.6	23.6	28.3	31.1	31.5	31.1	30.1	24.5	16.7	12.3
		MCWB	9.6	8.5	15.7	15.8	19.0	22.1	23.0	21.8	21.6	17.5	12.7	11.0
	2%	DB	6.4	6.5	14.5	19.9	25.8	29.2	29.7	28.4	27.1	21.3	13.9	8.7
		MCWB	5.3	4.4	10.2	13.6	18.4	20.8	21.6	21.0	20.4	16.7	11.6	7.5
	5%	DB	3.4	3.4	10.6	17.0	23.3	27.3	28.0	26.8	25.0	18.7	12.2	6.1
		MCWB	2.5	1.8	7.8	11.5	16.7	20.3	20.9	20.2	19.0	15.0	10.0	5.0
	10%	DB	1.1	1.2	7.2	14.0	20.7	25.1	26.4	25.4	22.7	16.4	9.8	3.7
		MCWB	0.4	0.0	4.6	9.2	14.9	19.0	19.9	19.5	18.0	13.6	8.0	2.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	8.5	15.8	17.6	21.0	23.7	24.8	23.9	23.2	20.2	14.2	11.0
		MCDB	10.3	10.1	20.6	21.9	26.2	29.0	29.4	28.0	27.4	22.4	15.6	11.7
	2%	WB	5.6	4.7	11.1	14.9	19.6	22.3	23.3	22.6	21.7	17.4	12.1	7.8
		MCDB	6.1	6.3	13.7	18.7	24.1	27.4	27.4	26.6	25.5	19.9	13.4	8.6
	5%	WB	2.6	1.7	8.0	12.6	17.9	21.1	22.2	21.6	20.2	15.6	10.0	5.0
		MCDB	3.4	3.0	11.0	15.9	22.0	25.5	26.1	25.2	23.3	18.1	11.7	5.9
	10%	WB	0.4	0.2	4.8	9.6	16.0	20.0	21.1	20.6	18.8	13.9	8.0	2.8
		MCDB	1.0	1.2	7.0	13.1	19.4	23.8	24.7	23.9	21.7	16.0	10.0	3.6
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.5	9.3	10.7	11.7	11.5	11.7	11.7	11.9	10.0	7.8	6.5
		MCDBR	9.2	10.3	14.1	16.5	15.2	14.1	13.3	13.7	14.8	13.8	11.4	9.1
		MCWBR	8.7	8.8	10.4	10.9	8.7	7.3	6.8	7.2	8.1	9.1	9.0	8.3
	5% WB	MCDBR	9.0	9.6	13.5	13.9	13.5	12.0	11.2	11.1	11.9	12.0	10.2	8.9
		MCWBR	8.8	8.6	10.7	10.5	8.6	7.1	6.6	6.7	7.3	8.7	8.7	8.6
Clear-Sky Solar Irradiance	taub	0.286	0.329	0.325	0.363	0.387	0.419	0.410	0.399	0.368	0.350	0.306	0.286	
	taud	2.371	2.198	2.394	2.334	2.312	2.239	2.281	2.301	2.382	2.418	2.503	2.432	
	Ebn at Noon	846	851	915	902	885	854	857	855	858	826	822	813	
	Edh at Noon	78	111	104	120	126	136	129	123	105	89	69	67	
All-Sky Solar Radiation	RadAvg	1.40	2.22	3.49	4.44	5.37	5.91	5.89	5.15	4.13	2.47	1.53	1.04	
	RadStd	0.17	0.26	0.35	0.50	0.53	0.40	0.41	0.26	0.39	0.23	0.20	0.13	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page